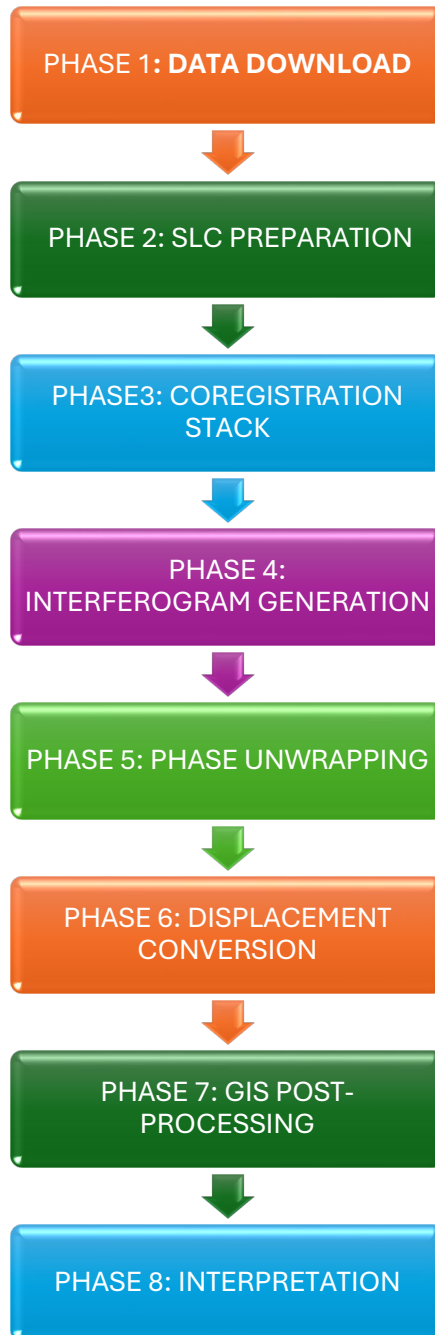


InSAR Time Series Subsidence Analysis for A10 Highway Netherlands (Sloterdijk to Rijksweg)

Technical Workflow Documentation

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Workflow



1. Sentinel-1 SLC Data Download

13 Sentinel-1 SLC images (January 1 – May 24, 2024) were downloaded from the **Copernicus Open Access Hub** (<https://scihub.copernicus.eu>).

Search Criteria:

- **Mission:** Sentinel-1A / Sentinel-1B
- **Product Type:** SLC (Single Look Complex)
- **Mode:** IW (Interferometric Wide)
- **Subswath:** IW3
- **Polarization:** VV
- **Relative Orbit:** 37 (descending)

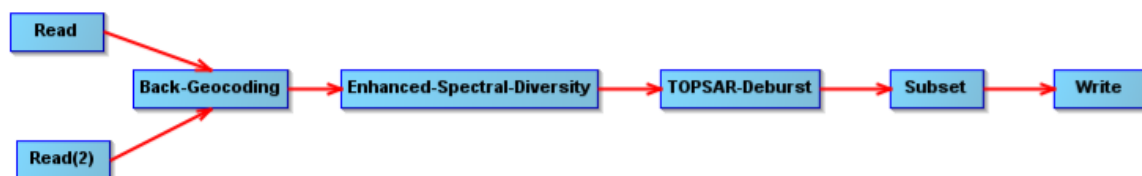
2. SLC PREPARATION

Original SLC → TOPS Split → Apply Orbit → Write

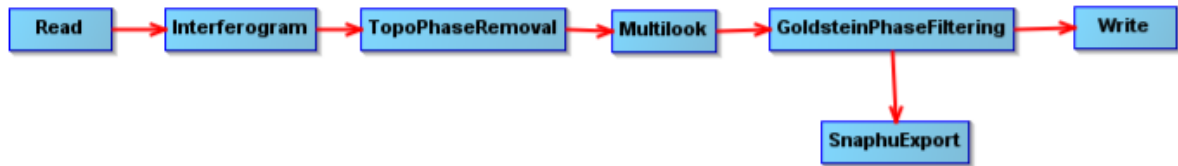


Repeat for all 13 SLC files (1 Master + 12 Slaves)

3. COREGISTRATION STACK



4. INTERFEROGRAM GENERATION



5. PHASE UNWRAPPING

Snaphu Export → SNAPHU → Snaphu Import

Output: Unwrapped phase

6. DISPLACEMENT CONVERSION

Phase to Displacement

Wavelength = 0.0554658 m (Sentinel-1)

Displacement (m) = $(\text{Phase} \times \lambda) / (4\pi)$

Terrain Correction

Map Projection: EPSG:28992 (RD New)

Output: GeoTIFF (32-bit float)

7. GIS POST-PROCESSING

Load GeoTIFF → A10 Centre line → 100m Buffer → Extract by Mask → Stack Profile

Time Series Analysis Graph from Python

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

# Load data
df = pd.read_excel(r'C:\Users\Lenovo\Documents\time series.xlsx', sheet_name='Sheet1')
# Convert chainage to kilometers
distance_km = df['Chainage'] / 1000
# Date labels
dates = ['Jan 13', 'Jan 25', 'Feb 6', 'Feb 18', 'Mar 1', 'Mar 13',
        'Mar 25', 'Apr 6', 'Apr 18', 'Apr 30', 'May 12', 'May 24']
# Colors: Green (early) -> Blue (middle) -> Red (late)
color_list = [
    '#006400', '#228B22', '#32CD32', '#7FFF00',
    '#0000CD', '#1E90FF', '#00BFFF', '#87CEEB',
    '#FF4500', '#FF0000', '#DC143C', '#8B0000'
]
# Create figure
plt.figure(figsize=(16, 10))
# Plot each date
for i, (date, color) in enumerate(zip(dates, color_list)):
    displacement_mm = df.iloc[:, i + 1] * 1000
    plt.plot(distance_km, displacement_mm, color=color, linewidth=2.2, alpha=0.9, label=date)
# Reference line at 0
plt.axhline(y=0, color='black', linestyle='--', linewidth=1.5, alpha=0.7)
# Title
plt.title('InSAR Time Series Analysis\nA10 Highway Subsidence Profile (Jan 1 - May 24, 2024)',
        fontsize=16, fontweight='bold', pad=20)
# Labels
plt.xlabel('Chainage along A10 Highway (km)', fontsize=14, fontweight='bold')
plt.ylabel('Displacement (mm)', fontsize=14, fontweight='bold')
# Location annotation (top left, italic, no "Location:" prefix)
plt.annotate('Netherlands, Amsterdam to Amstel Park',
        xy=(0.02, 0.98), xycoords='axes fraction', fontsize=11,
        ha='left', va='top', style='italic', alpha=0.7,
        bbox=dict(boxstyle='round', facecolor='f0f0f0', alpha=0.8, edgecolor='none'))
# Axis limits
plt.ylim(-90, 30)
plt.xlim(0, 5.2)
plt.xticks(np.arange(0, 5.5, 0.3), fontsize=10)
plt.yticks(fontsize=11)
# Grid
plt.grid(True, alpha=0.3, linestyle='--', linewidth=0.8)
# Legend outside
plt.legend(loc='upper left', bbox_to_anchor=(1.01, 1), fontsize=10,
        title='Date', title_fontsize=11, framealpha=0.95, ncol=1, fancybox=True)
# Background
plt.gca().set_facecolor('#fafafa')
# Color ramp explanation (bottom right)
plt.annotate('Color: Green (Jan-Feb) → Blue (Mar-Apr) → Red (Apr-May)',
        xy=(0.98, 0.02), xycoords='axes fraction', fontsize=9,
        ha='right', va='bottom', style='italic', alpha=0.6)
# Adjust layout
plt.tight_layout()
# Save
plt.savefig(r'C:\Users\Lenovo\Documents\A10_Highway_InSAR_Timeseries.png',
        dpi=300, bbox_inches='tight', facecolor='white')
# Show
plt.show()
print("✅ Graph saved: A10_Highway_InSAR_Timeseries.png")
print("📍 Netherlands, Amsterdam to Amstel Park (top left)")
print("📅 Jan 1 - May 24, 2024")
print("🌈 Green (early) → Blue (middle) → Red (late)")
```

8. INTERPRETATION

- Negative displacement = Subsidence (sinking)
- Positive displacement = Uplift (rising)
- Hotspot identification at chainage ~2,356 m (85 mm subsidence)
- Color ramp: Green (Jan-Feb) → Blue (Mar-Apr) → Red (Apr-May)

Key Parameters Summary

Parameter	Value
Master date	2024-03-13
Number of slaves	12
Total images	13
Polarization	VV
Subswath	IW3
DEM	SRTM 3 arc-sec
Goldstein alpha	0.5
Multilook	4×4
Wavelength	0.0554658 m
Output CRS	EPSG:28992 (RD New)

Software Used

Software	Purpose
SNAP 9.x	SAR preprocessing, interferogram generation
SNAPHU 1.4.2	Phase unwrapping
ArcGIS Pro	GIS analysis, profile extraction
Python 3.12	Time series matrix, visualization

Subset Polygon (AOI)

Text POLYGON((4.788861 52.402344, 4.890706 52.391258, 4.892422 52.323042, 4.789272 52.328783, 4.788861 52.402344))

Results Summary

Metric	Value
Maximum subsidence	~85 mm
Hotspot chainage	~2,356 m
Time period	144 days (Jan 1 – May 24)
Subsidence rate	~17 mm/month (~204 mm/year)

Limitations

This analysis has some limitations. Vegetated areas like trees and parks along the highway have low radar coherence, so displacement values in those zones may be noisy or unreliable. Atmospheric effects such as water vapor were not corrected, which can introduce seasonal errors. The results show line-of-sight movement only, not true vertical or horizontal motion. No ground survey data was available to verify the InSAR results. The 85 mm hotspot at chainage 2,356 meters falls on a vegetated area, so this value should be interpreted with caution.